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Using CMSIS-DSP with MCUXpresso SDK and IDE

Documento creado por [Carlos_Mendoza](#)  el 29-sep-2017

 Versión 1

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Using CMSIS-DSP with MCUXpresso SDK and IDE

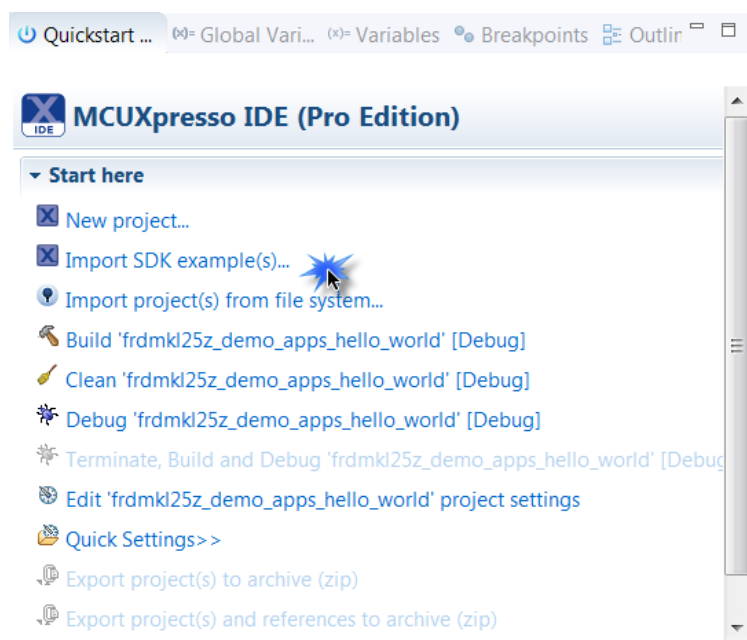
Follow these steps to link the CMSIS-DSP library to a MCUXpresso SDK 2.x project using the MCUXpresso IDE. The steps described in the document were done using the MKL25Z MCU like the one in the FRDM-KL25Z board, but the same principles are applicable to any Kinetis MCU.

Please make sure you have already created and installed the corresponding MCUXpresso SDK package to the MCUXpresso IDE, you can use following links as reference:

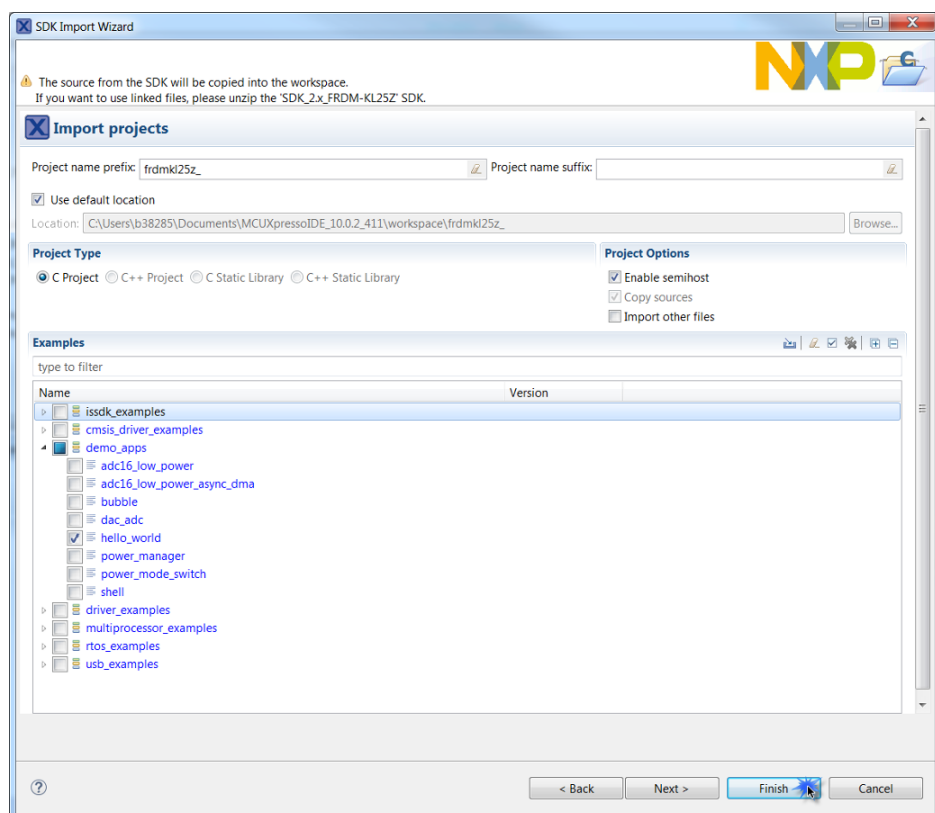
-  [Getting Started with MCUXpresso and FRDM-K64F](#)
-  [Generating a downloadable MCUXpresso SDK v.2 package](#)

Creating a MCUXpresso SDK 2.x Project:

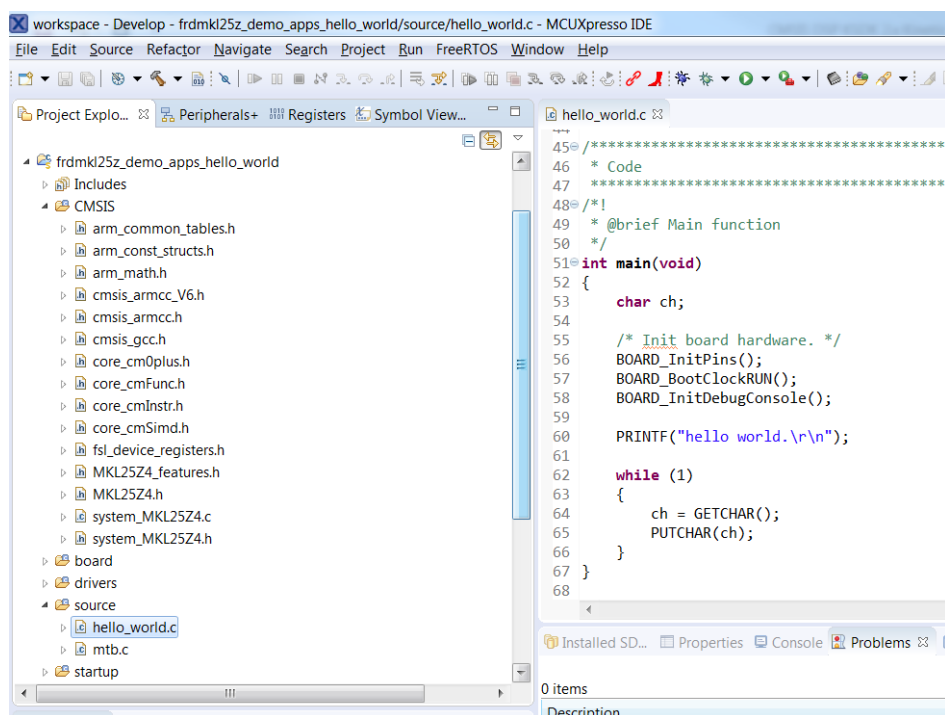
1) Click on the **Import SDK example** option from the Quickstart menu:



2) For this demonstration the **Hello World** example is used:

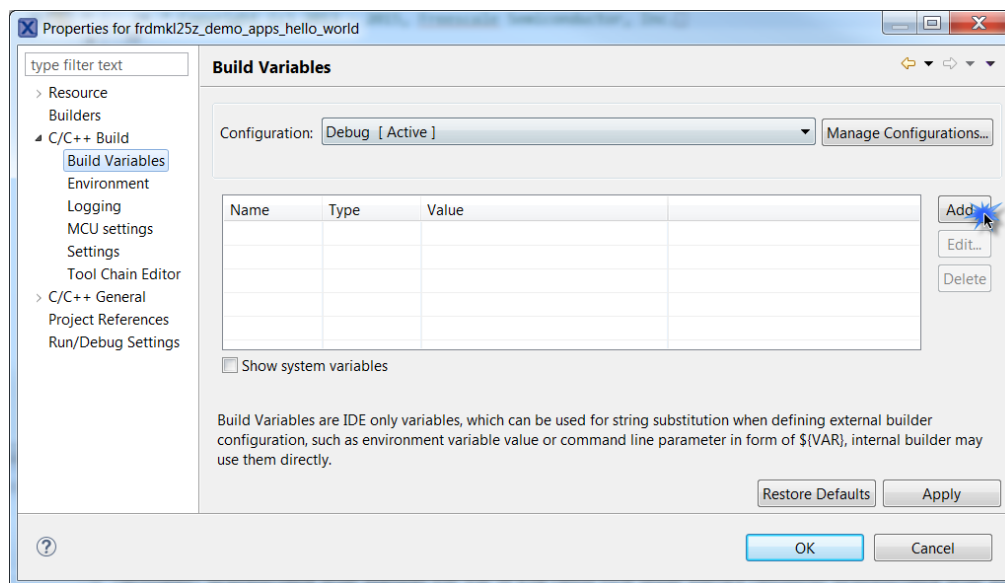


3) The new project should now appear on your workspace:

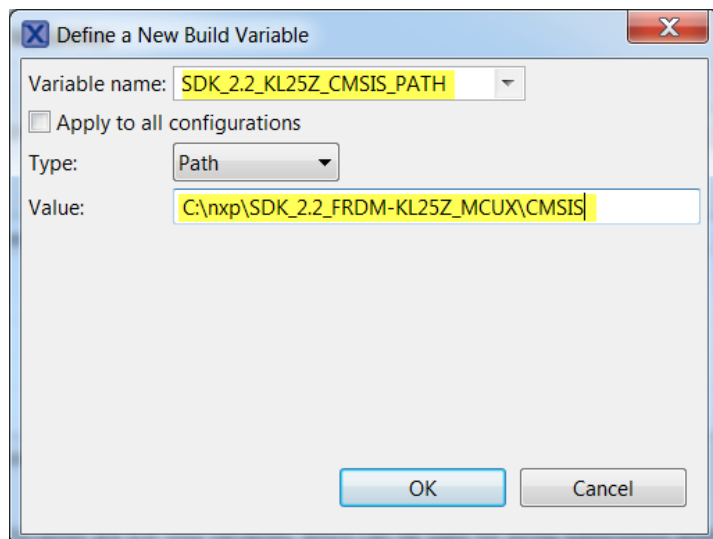


Linking CMSIS-DSP Library:

1) The first step is to create a build variable that will be used to specify the path of the DSP library. Go to **Project > Properties** and under **C/C++ Build** select **Build Variables** and click on **Add**:



2) A new window will open, specify the **name** of the build variable, its type and value, the Value is the **location** of your CMSIS folder:

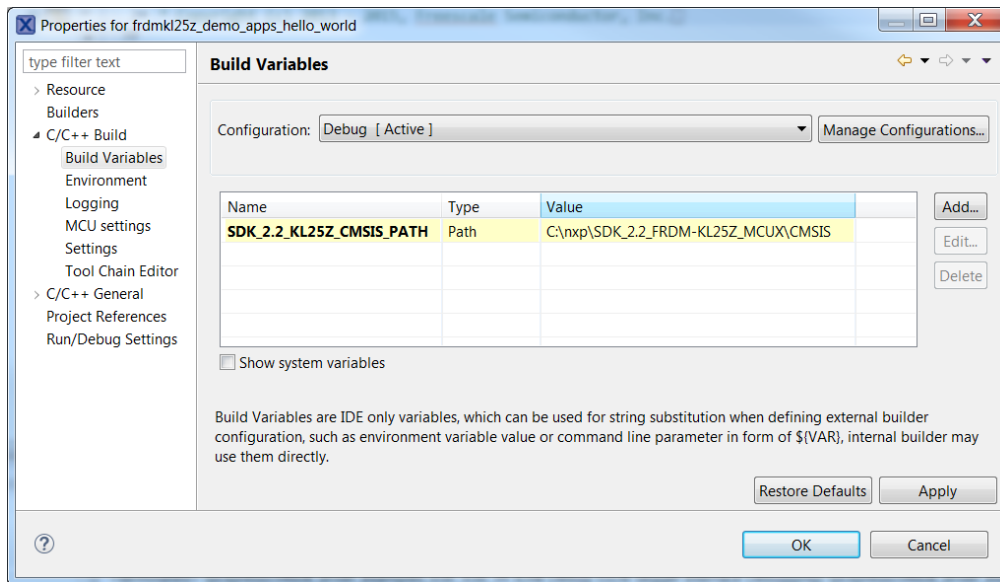


Variable name: `SDK_2.2_KL25Z_CMSIS_PATH`

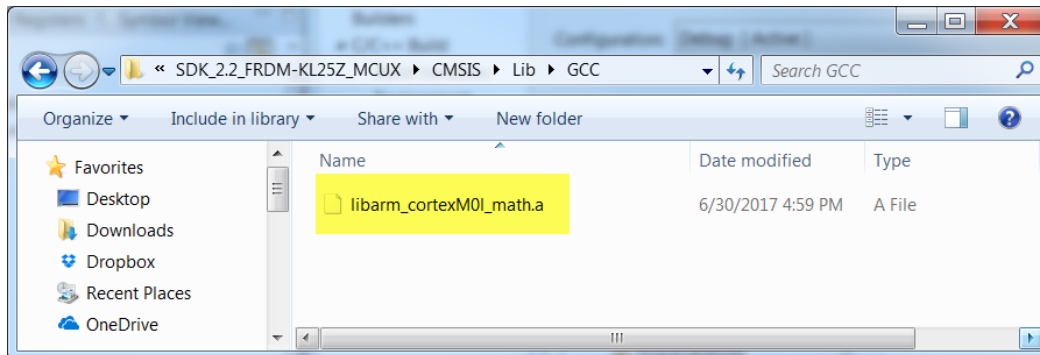
Value: `C:\nxp\SDK_2.2_FRDM-KL25Z_MCUX\CMSIS`

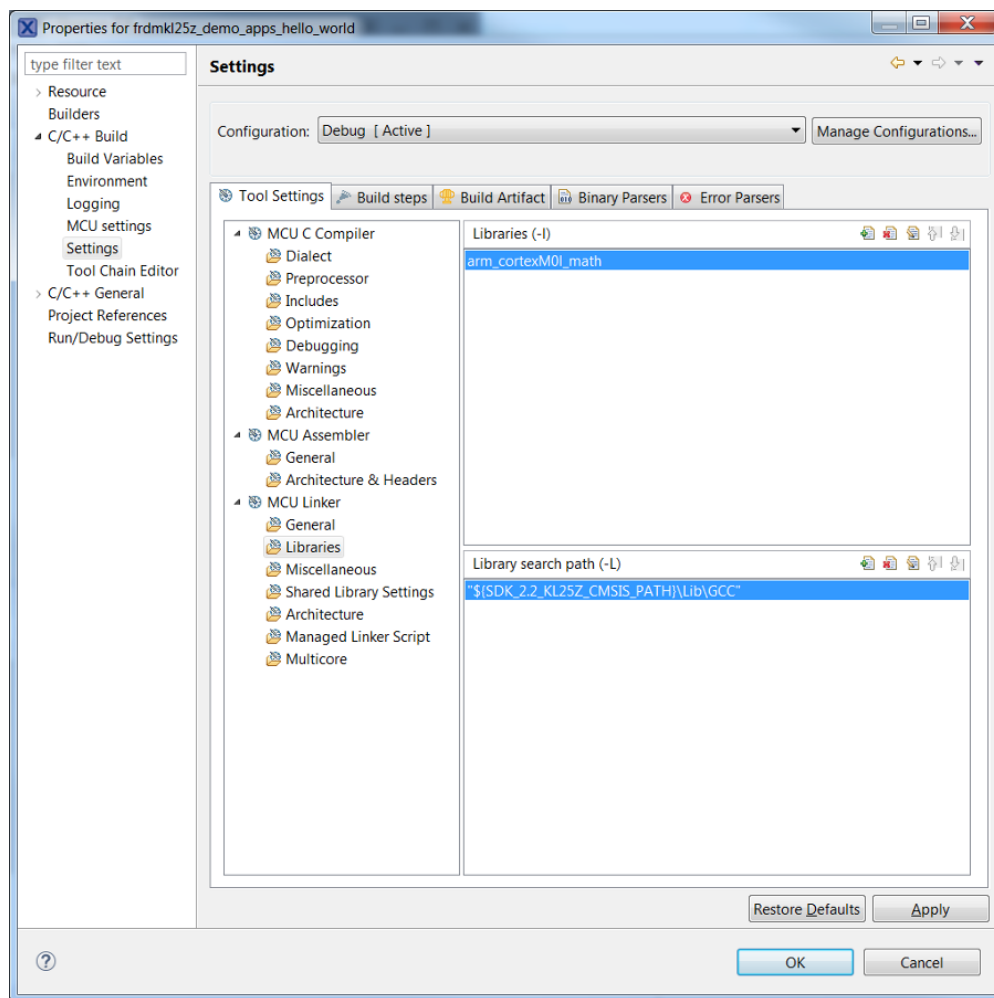
NOTE: The `SDK_2.2_FRDM-KL25Z` package was previously unzipped to the `C:\nxp` folder

3) The new variable should be listed as in the image below, click on **Apply**:



4) Go to **C/C++ Build > Settings > MCU Linker > Libraries** and specify the precompiled library to be used and its path:



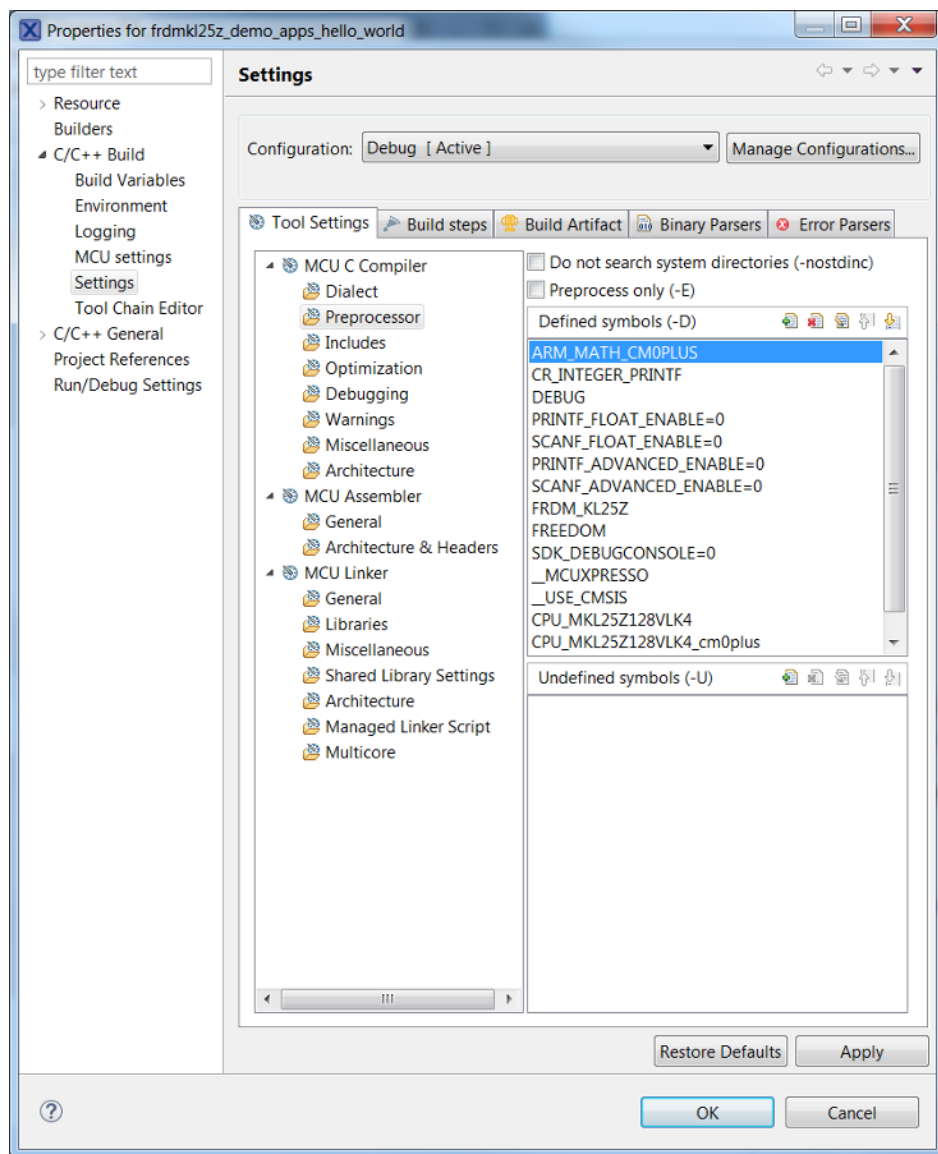


Library name: arm_cortexM0l_math. The M denotes the ARM core, while the 'l' means 'little endian'.

Path: \$(SDK_2.2_KL25Z_CMSIS_PATH)\Lib\GCC

5) Now go to **C/C++ Build > Settings > MCU C Compiler > Preprocessor** and specify the following macro:

ARM_MATH_CM0PLUS: Tells the CMSIS library which ARM Cortex core I'm using.



Importing DSP example source files:

1) For this project the “**FIR Lowpass Filter**” example will be used, it can be found on the following path:

`${SDK_2.2_KL25Z_CMSIS_PATH}\DSP_Lib\Examples\arm_fir_example\ARM`

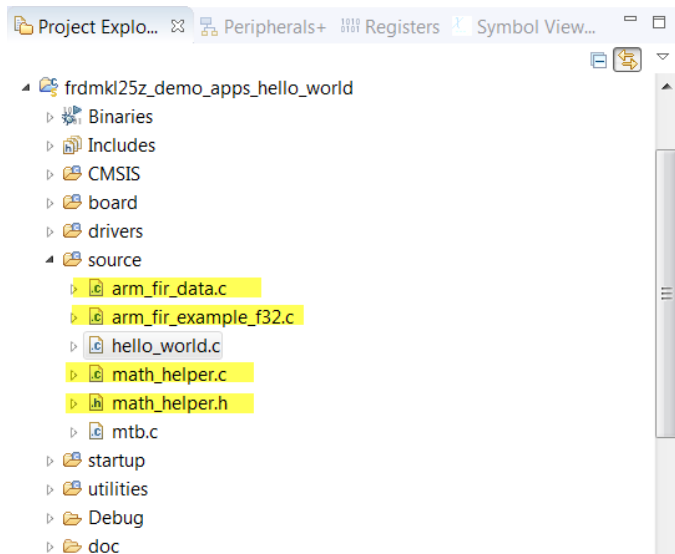
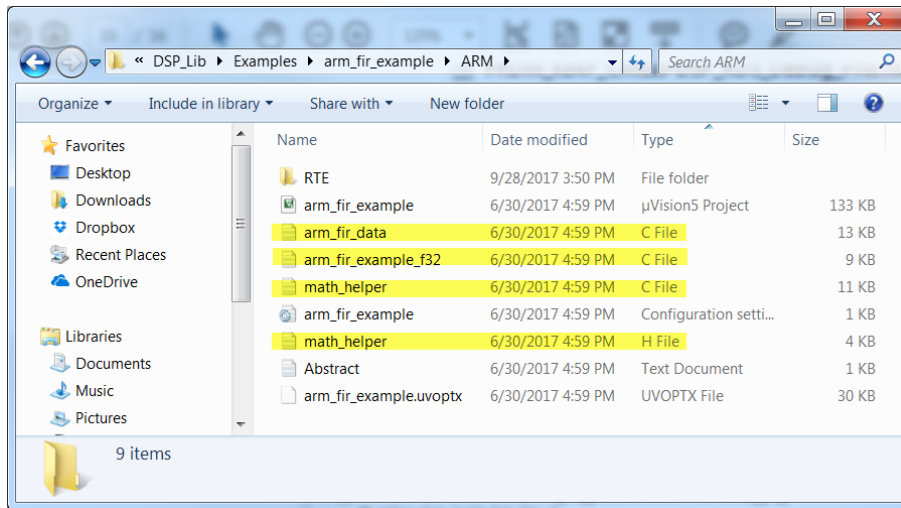
2) The first step is to **copy** the source files of the example to the project, the files that need to be copied are:

`arm_fir_example_f32.c`

`arm_fir_data.c`

`math_helper.c`

`math_helper.h`



3) The next step is to copy the SDK include files and initialization functions from the hello_world.c file to the FIR example file:

```
#include "fsl_device_registers.h"
#include "fsl_debug_console.h"
#include "board.h"
#include "pin_mux.h"
```

```
BOARD_InitPins();
BOARD_BootClockRUN();
BOARD_InitDebugConsole();
```

```

hello_world.c  arm_fir_example_f32.c
20 * Copyright (c) 2012-2015 Freescale Semiconductor, Inc.
30 ftdmki25z_demo_apps_hello_world/source/hello_world.c
31 #include "fsl_device_registers.h"
32 #include "fsl_debug_console.h"
33 #include "board.h"
34 #include "pin_mux.h"
35
36 /**
37  * Code
38  */
39 /*
40  * @brief Main function
41  */
42 int main(void)
43 {
44     char ch;
45
46     /* Init board hardware. */
47     BOARD_InitPins();
48     BOARD_BootClockRUN();
49     BOARD_InitDebugConsole();
50
51     PRINTF("hello world.\r\n");
52

```

```

hello_world.c  arm_fir_example_f32.c
117
118 /** \example arm_fir_example_f32.c
119  */
120
121 /** -----
122  ** Include Files
123  ** -----
124
125 #include "arm_math.h"
126 #include "math_helper.h"
127
128 #include "fsl_device_registers.h"
129 #include "fsl_debug_console.h"
130 #include "board.h"
131 #include "pin_mux.h"
132

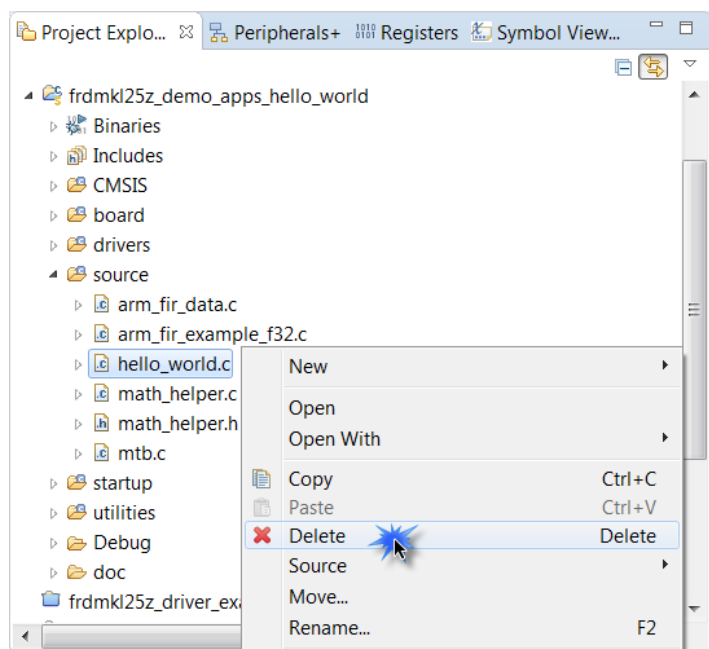
```

```

hello_world.c  arm_fir_example_f32.c
183
184 /** -----
185  * FIR LPF Example
186  * -----
187
188 int32_t main(void)
189 {
190     uint32_t i;
191     arm_fir_instance_f32 S;
192     arm_status status;
193     float32_t *inputF32, *outputF32;
194
195     /* Initialize input and output buffer pointers */
196     inputF32 = &testInput_f32_1kHz_15kHz[0];
197     outputF32 = &testOutput[0];
198
199     /* Init board hardware. */
200     BOARD_InitPins();
201     BOARD_BootClockRUN();
202     BOARD_InitDebugConsole();
203

```

4) Finally the `hello_world.c` file can be deleted from the project:



5) Now you should be able to compile and debug the project.

Related links:

- [Generating a downloadable MCUXpresso SDK v.2 package](#)
- [Getting Started with MCUXpresso and FRDM-K64F](#)
- [Adding CMSIS-DSP Library to a KSDK 2.x project in Kinetis Design Studio](#)

1 persona indicó que le resultó útil

ADJUNTOS

RESULTADOS

Útil(1)

Visibilidad:  MCUXpresso Software and Tools • 2248 vistas

Modificado por última vez el 02-oct-2017 11:10

Clasificaciones:

Calificación promedio del usuario



(0 Clasificaciones)

1 comentario



Tak Ming Marco Pang

03-oct-2017 19:26

Hey! Thank you for the guide!

I have a question about using CMSIS-DSP,

I'm running Kinetis Design Studio 3.2, MQX 4.2 and SDK 1.x project. There is SDK2.x but it doesn't seem to support K70F120M board.

Are you instructions similar in that I can just link it like you did?

Sorry if this is the wrong place to ask.

 Acciones ▾

 Me gusta • 1

Contenido relacionado

-  [Getting Started with MCUXpresso and FRDM-K64F](#)
-  [How to Flash Two or More Binary Files Using MCUXpresso IDE](#)
-  [How to Switch C/C++ Library in MCUXpresso IDE](#)
-  [Quick Start Guide – Using MCUXpresso SDK with PINs&CLOCKs Config Tools](#)
-  [FAQ: MCUXpresso Software and Tools](#)

Contenido recomendado

-  [I am having an issue regarding running my python file on the Bluebox2.0. Can someone help me address the issue?](#)
-  [Enabling boot console in Yocto](#)
-  [Autonomous Clock Trimming SW example for MagniV devices](#)
-  [Redlib documentation missing?](#)
-  [S32DS creating position independent code for AN12323 problem.](#)

Enlaces entrantes

-  [Re: CMSIS DSP library porting in LPC54608J512BD208](#)
-  [Re: CMSIS DSP library porting in LPC54608J512BD208](#)
-  [CMSIS DSP library porting in LPC54608J512BD208](#)
-  [Re: LPC55S69 PowerQuad Example](#)
-  [Link DSP Library](#)

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